

Phase 2: Interview Orchestrator & Engine Design

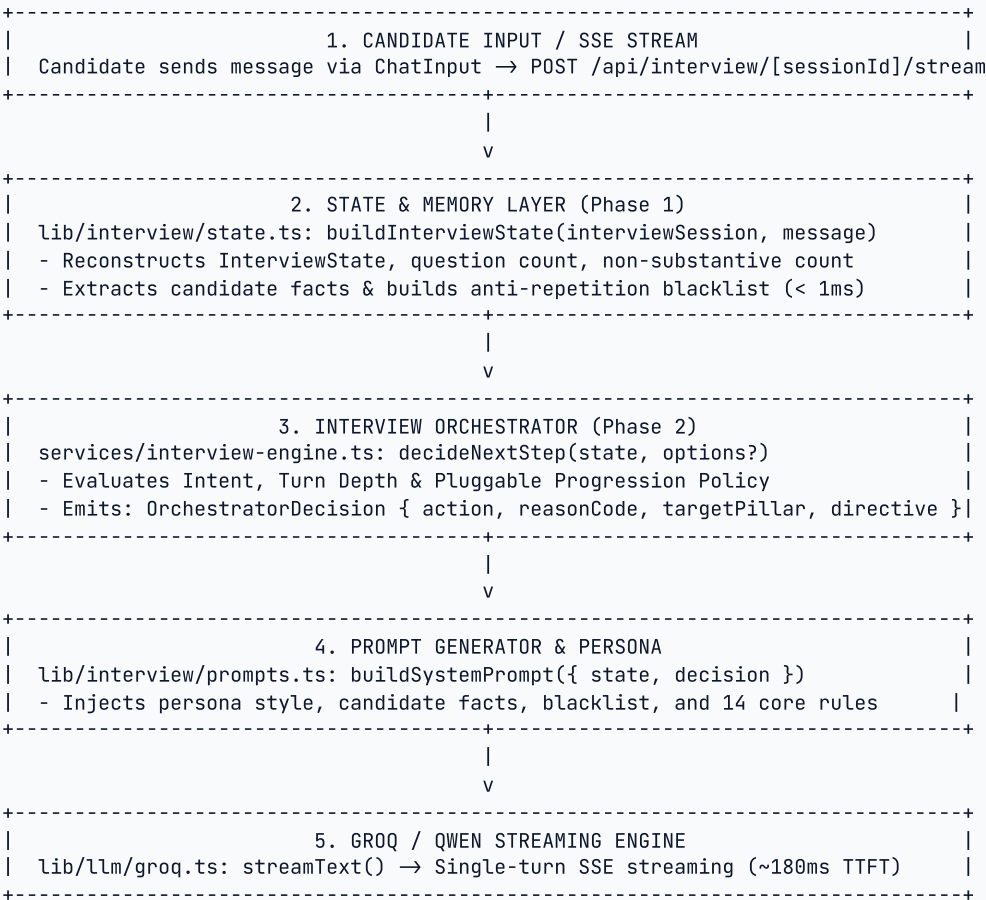
Deterministic Decision Plane, Reason Codes & Pluggable Progression Architecture

REVIEW STATUS	CORE TARGET	STATE LAYER	DATE
Approved Design with Feedback	services/interview-engine.ts	lib/interview/state.ts (Phase 1)	September 29, 2026

Three Core Design Enhancements Incorporated:

- 1. **Explicit Reason Codes:** Added `reasonCode` to `OrchestratorDecision` for complete observability and telemetry.
- 2. **Pluggable Progression Policy:** Structured turn-count matrix as the baseline policy, allowing Phase 3 evaluation metrics to seamlessly influence decisions without architectural rewrites.
- 3. **Strict Clarification Directive:** `ANSWER_CLARIFICATION` only generates the directive instruction; the LLM synthesizes the natural technical explanation.

1. System Architecture & Request Lifecycle



2. Enhanced Orchestrator Contracts

Input Contract

```
export interface ProgressionPolicyOptions {  
  turnsPerPillarTarget?: number; // Baseline default: 2 substantive turns  
  evaluationScoreOverride?: number; // Phase 3 hook: evaluation mastery score  
}  
  
export interface OrchestratorInput {  
  state: InterviewState;  
  candidateMessage?: string;  
  isOpening?: boolean;  
  options?: ProgressionPolicyOptions;  
}
```

Output Contract (`OrchestratorDecision` with `reasonCode`)

```
export type InterviewAction =  
  | "OPEN_INTERVIEW" // Turn 0: Persona intro & background request  
  | "ACKNOWLEDGE_INTRO" // Turn 1: Anchor to candidate's intro & ask Pillar 1  
  | "HANDLE_GREETING" // Greeting: brief warm phrase + repeat question  
  | "PROMPT_FOR_SUBSTANCE" // Non-substantive 1: Reprompt patiently without validation  
  | "PIVOT_AWAY" // Non-substantive 2+: Release topic & switch pillar  
  | "ANSWER_CLARIFICATION" // Candidate asked question: directive for LLM to answer in 1-2 sentences  
  | "HANDLE_META_REQUEST" // Candidate meta-instruction: put candidate in hot seat  
  | "DRILL_DOWN" // Substantive 1: probe tradeoff, mechanics, concurrency  
  | "CHALLENGE_TRADEOFF" // Substantive 2: challenge failure modes & constraints  
  | "ADVANCE_PILLAR" // Substantive 3+: transition to next domain pillar  
  | "CONCLUDE_ROUND"; // Wrap up requested or round complete: emit [SESSION_COMPLETED]  
  
export type DecisionReasonCode =  
  | "INITIAL_OPENING"  
  | "CANDIDATE_REQUESTED_END"  
  | "GREETING_DETECTED"  
  | "CANDIDATE_META_INSTRUCTION"  
  | "CANDIDATE_ASKED_CLARIFICATION"  
  | "NON_SUBSTANTIVE_REPROMPT"  
  | "NON_SUBSTANTIVE_PIVOT_LIMIT"  
  | "CANDIDATE_INTRO_RECEIVED"  
  | "SUBSTANTIVE_PROBE_DEPTH"  
  | "SUBSTANTIVE_CHALLENGE_TRADEOFF"  
  | "PILLAR_PROGRESSION_ADVANCE"  
  | "ALL_PILLARS_EXHAUSTED";  
  
export interface OrchestratorDecision {  
  action: InterviewAction;  
  reasonCode: DecisionReasonCode;  
  targetPillarIndex: number;  
  targetPillarName: string;  
  consecutiveTurnsOnPillar: number;  
  isTerminalTurn: boolean;  
  directiveInstruction: string;  
  groundingKeywords: string[];  
}
```

3. Deterministic Action Decision Matrix with Reason Codes

Condition	Action	Reason Code	Orchestrator Directive
Opening Turn	OPEN_INTERVIEW	INITIAL_OPENING	Introduce persona name/role, ask candidate for background. No technical questions.
End Intent	CONCLUDE_ROUND	CANDIDATE_REQUESTED_END	1-2 sentence closing thank you. Append [SESSION_COMPLETED] . No further questions.
Greeting	HANDLE_GREETING	GREETING_DETECTED	1 short phrase greeting. Immediately continue the SAME question.
Meta Request	HANDLE_META_REQUEST	CANDIDATE_META_INSTRUCTION	Do not answer yourself. Ask candidate directly about X to keep them in the hot seat.
Clarification	ANSWER_CLARIFICATION	CANDIDATE_ASKED_CLARIFICATION	Answer candidate's technical query concisely in 1-2 sentences, then prompt them to proceed.
Non-Answer 1	PROMPT_FOR_SUBSTANCE	NON_SUBSTANTIVE_REPROMPT	Do not affirm ("Got it"). Patiently reprompt for concrete details and tools.
Non-Answer 2+	PIVOT_AWAY	NON_SUBSTANTIVE_PIVOT_LIMIT	Release topic naturally ("All good, let's look at..."). Advance to next pillar.
Candidate Intro	ACKNOWLEDGE_INTRO	CANDIDATE_INTRO_RECEIVED	Acknowledge 1 specific project/tool from intro. Ask opening question from Pillar 1.
Substantive 1	DRILL_DOWN	SUBSTANTIVE_PROBE_DEPTH	Anchor to specific detail. Probe internal mechanics, data structures, or trade-offs.
Substantive 2	CHALLENGE_TRADEOFF	SUBSTANTIVE_CHALLENGE_TRADEOFF	Challenge edge cases, high concurrency, or failure scenarios.
Substantive 3+	ADVANCE_PILLAR	PILLAR_PROGRESSION_ADVANCE	Transition to the next unvisited domain pillar in the track.

4. Clarification Boundary & Pluggable Progression Architecture

Strict Clarification Boundary (`ANSWER\_CLARIFICATION`)

To preserve clean architectural boundaries:

- The **Orchestrator** identifies that the candidate asked a question and outputs a directive: *"The candidate asked a clarification regarding the architecture. Answer knowledgeably in 1-2 sentences, then resume the interview scenario."*
- The **Orchestrator NEVER hard-codes technical answers**.
- The **LLM synthesizes the natural technical explanation** within the persona's voice and background.

Pluggable Progression Architecture (Phase 2 → Phase 3 Evolution)

Phase 2 Baseline → Phase 3 Evaluation Hooks

In Phase 2, the progression policy utilizes the turn-count heuristic (2 substantive turns per pillar). The method `shouldAdvancePillar(state, options)` is modularized so that in Phase 3, real-time rubric scores (e.g. candidate demonstrated mastery after 1 turn or requires remediation) can dynamically adjust the threshold without refactoring the orchestrator core.

5. Phase 2 Implementation Boundaries

File Path	Target Status	Implementation Scope
services/interview-engine.ts	IMPLEMENT	Implement InterviewOrchestrator class with decideNextStep(state: InterviewState) and reasonCode .
lib/interview/prompts.ts	ENHANCE	Update buildSystemPrompt to consume OrchestratorDecision directly.
app/api/interview/[sessionId]/stream/route.ts	WIRE	Wire InterviewOrchestrator.decideNextStep into the live turn streaming route.
lib/auth/* & middleware.ts	DO NOT MODIFY	Candidate NextAuth, isolated Admin HMAC auth, route protection.
prisma/schema.prisma	DO NOT MODIFY	Zero database schema modifications.
app/interview/[sessionId]/page.tsx	DO NOT MODIFY	UI layout, design system, theme tokens, SSE reader.